

Jeonggyu Hwang

Graduate Student

EDUCATION

March 2025 - Present Combined Master-Doctor Student in
School of Electrical & Electronic Engineering,
Yonsei University

- Supervisor: Prof. Lee, Taeyoon Ph.D.
- Research topic: VO₂ Insulator

RESEARCH EXPERIENCE

January 2025 - Present Graduate Research Assistant
NanoBio Device Lab. (NBDL), Yonsei University

ACADEMIC EXPERIENCE

February 2026 - June 2026 Teaching Assistant (TA)
Physical Electronics (EEE3210), Spring 2026

January 2026 - Present Administrative Assistant
School of Electrical & Electronic Engineering,
Yonsei University

PUBLICATIONS

Conference Hwang, J., & Lee, T. (2026). Study of germanium sulfide films with different nitrogen doping levels for ovonic threshold switches. Poster presentation at GCIM 2026 & MRS-K Spring Meeting.

HONORS & AWARDS

November 2025 Priority Selection for Yonsei Graduate school

April 2026 Incoming Excellence Scholarship for
School of Electrical & Electronic Engineering,
Yonsei University

PERSONAL INFO

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E-mail

jeonggyuhwang04@gmail.com

Portfolio

<https://developer-jg.github.io/>

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RESEARCH INTERESTS

- Semiconductor Device Physics
- Ovonic Threshold Switches
- Oxide Electronics
- Thin Film Growth and Interface Engineering

EDUCATION

Yonsei University

Combined Master-Doctor Student

School of Electrical & Electronic Engineering, College of Engineering

Seodaemun-gu, Seoul, South Korea

March 2026 - Present

Gachon University

Undergraduate Student

Major in Semiconductor Display, Department of Semiconductor Engineering,

College of Semiconductor

- Bachelor of Engineering (B.E.)
- GPA (Overall): 4.33 / 4.50 (Major GPA: 4.37 / 4.50)
- Honors: Cum Laude

Seongnam-si, Gyeonggi-do, South Korea

March 2023 - February 2026

Seoul National University

Undergraduate Exchange Student

Department of Materials Science and Engineering, College of Engineering

Gwanak-gu, Seoul, South Korea

August 2024 - February 2025

PUBLICATIONS

Conference:

- [2] **Hwang, J.**, & Lee, T. (2026). Study of germanium sulfide films with different nitrogen doping levels for ovonic threshold switches. Poster presentation at GCIM 2026 & MRS-K Spring Meeting.
- [1] **Hwang, J.**, & Jeon, J. (2025). Exploring the theoretical limits of efficiency in multilayer solar cells. Poster presented at the 32nd Korean Conference on Semiconductors (KCS), Korea.

Pre-Print:

- [3] **Hwang, J.** (2025). Estimation of Solar Spectral Irradiance Using Meteorological Data and Analysis of Optimal Conditions for Solar Power Generation. *arXiv preprint arXiv:2504.08008*.
- [2] **Hwang, J.** (2024). Theoretical analysis of doping concentration gradients on solar cell performance. *arXiv preprint arXiv:2407.12245*.
- [1] **Hwang, J.** (2024). Exploring the theoretical limits of efficiency in multilayer solar cells. *arXiv preprint arXiv:2404.14930*.

ACADEMIC EXPERIENCE

Yonsei University

Graduate Researcher

School of Electrical & Electronic Engineering, College of Engineering

- NanoBio Device Lab. (NBDL)

Seodaemun-gu, Seoul, South Korea

January 2026 - Present

- PI: Prof. Lee, Taeyoon
- Research topic: TBD

Undergraduate Researcher

June 2025 - July 2025

Department of Physics, College of Science

- Center for Bandstructure Engineering
- PI: Prof. Kim, Keunsu
- Research topic: 2D Materials and Solid-State Physics

POSTECH

Pohang-si, Gyeongsangbuk-do, South Korea

Pohang University of Science and Technology

Undergraduate Researcher

February 2024

Department of Chemical Engineering

- FIMDEM Lab. - Functional Inorganic Materials Design & Manufacturing
- PI: Prof. Son, Jaesung
- Research topic: Solution-Processed Thin Films and Thermoelectric Device Materials

Korea University

Seongbuk-gu, Seoul, South Korea

Undergraduate Researcher

July 2023 - August 2023

Department of Integrative Energy Engineering, College of Engineering

Graduate School of Energy and Environment

- Electrochemistry Lab. - Solarrino
- PI: Prof. Jun, Yongseok
- Research topic: Perovskite Solar Cells

INDUSTRY EXPERIENCE

Philoptics Co., Ltd

Osan-si, Gyeonggi-do, South Korea

Engineer

March 2024 - May 2025

Semiconductor Department, R&D Division

- Development of Advanced Glass Singulation Processing Technology
- Development of Next-Generation ABF Drilling Core Technology
- 2024 Advanced Technology Development Project

AWARDS AND HONORS

Incoming Excellence Scholarship, Yonsei University

Graduation with Honors, Gachon University

Selected for Priority Admission, Yonsei University

Academic Excellence Scholarship, Gachon University

Undergraduate Poster Presentation Award,

32nd Korean Conference on Semiconductor

LEADERSHIP & ACTIVITY

Research Service & Leadership

The Gachon Research Society

Honorary Member

February 2025 - Ongoing

Chairperson of the Acting Executive & Steering Committee for

the 2025 Delegates Joint Meeting

February 2025 - November 2025

Executive Adviser

December 2024 - February 2025

Founding President & 1st President

February 2024 - November 2024

- Contributed to improving language proficiency and fostering academic development

through reading and debating multilingual papers and books.

- Facilitated information exchange within the university and strengthened networking opportunities among student researchers.

Academic Service & Leadership

Teaching Assistant (TA)

Physical Electronics (EEE3210), Spring 2026 February 2026 - June 2026 (Expected)
School of Electrical & Electronic Engineering, Yonsei University

Administrative Assistant

School of Electrical & Electronic Engineering, Yonsei University January 2026 - Present

Executive Member

Student Association March 2023 - February 2024
Major in Semiconductor Display, Gachon University